

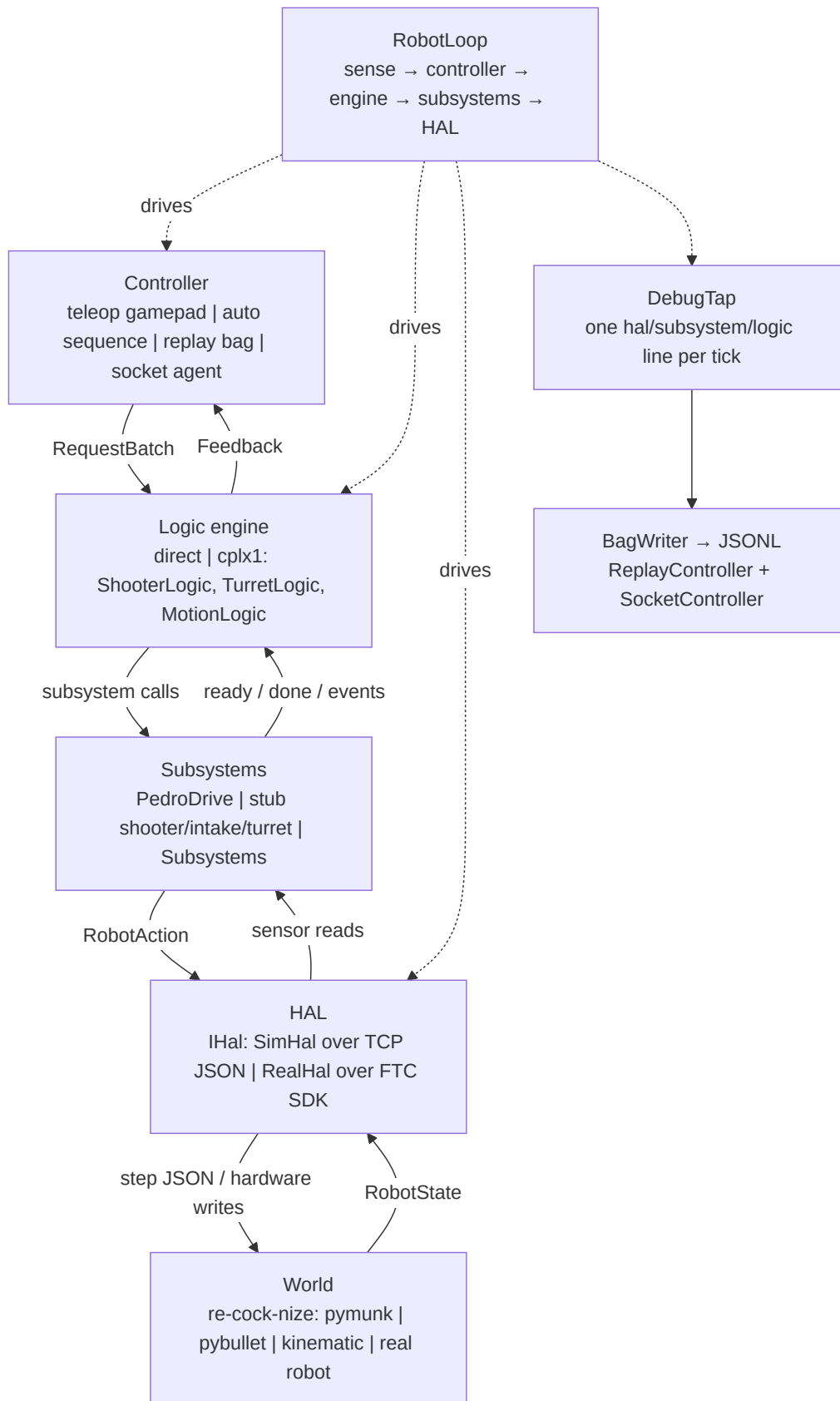


Figure 1: Diagram 1

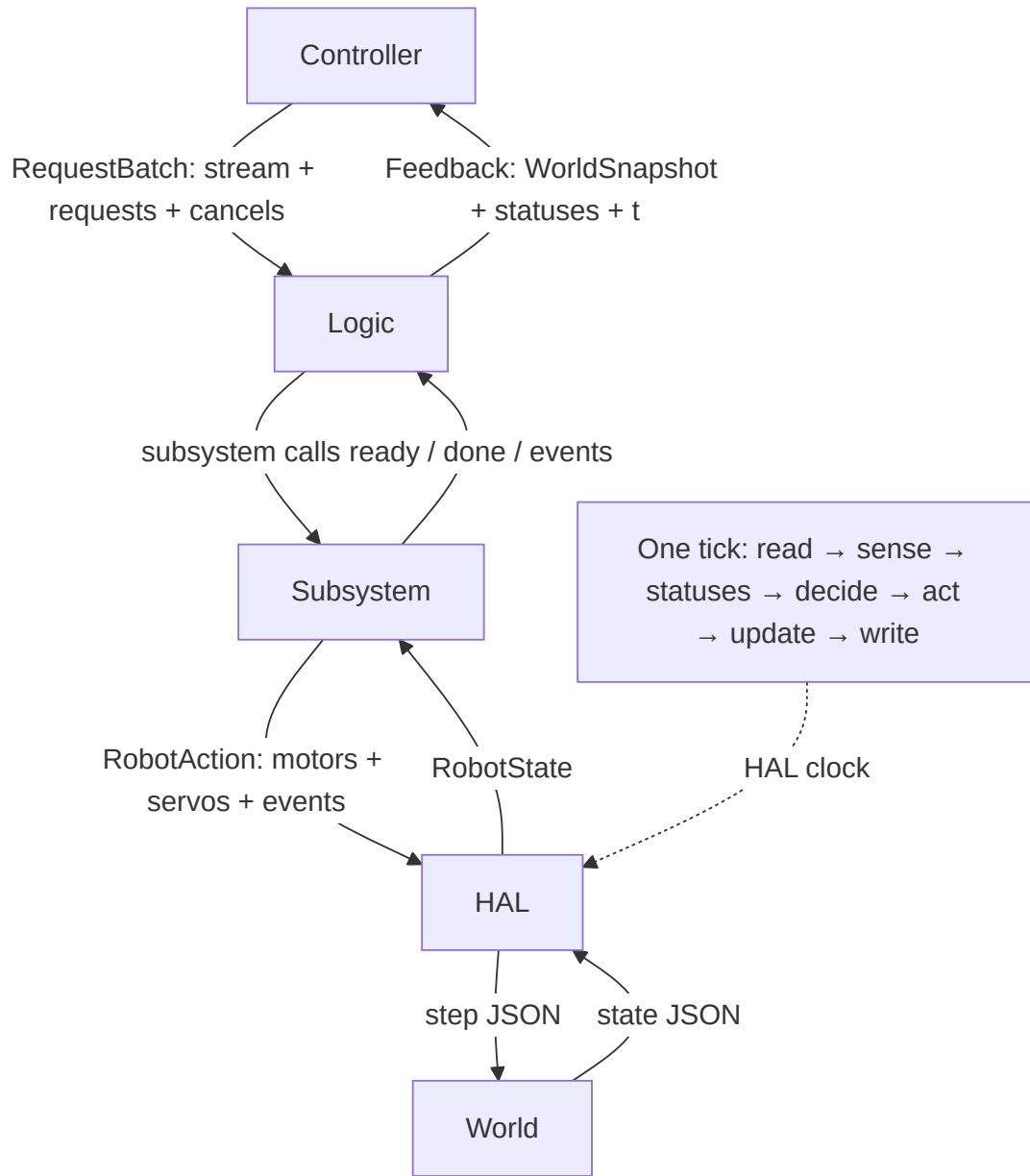


Figure 2: Diagram 2

Variant C – Follow one request

The examples use the current `robot-code dev-phase-1.1` head `dc4da66` and `re-cock-nize dev-phase-1.1` head `518bf9c`. IDs are illustrative values from the real allocators (teleop starts at 10000; `SequenceRunner` starts at 1).

1. SHOOT(3) from RB to DONE

1.1 Press, batch, and sense

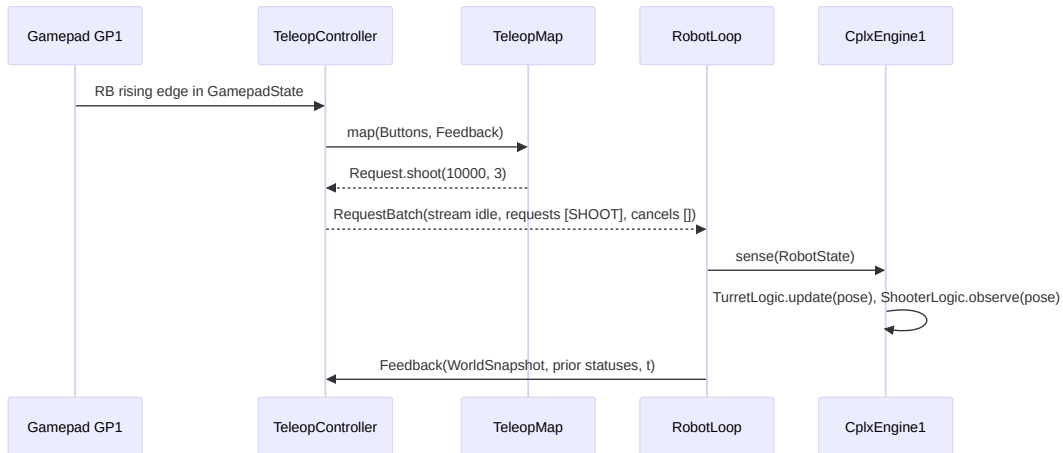


Figure 3: Diagram 3

At this edge the exact Java record is `Request(10000, SHOOT, double[]{3.0}, null)` inside `RequestBatch(RequestStream.idle(), [request], new int[0])`.

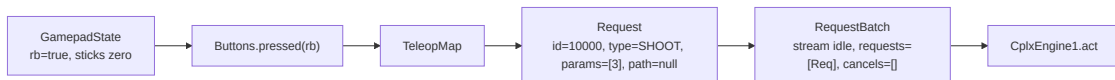


Figure 4: Diagram 4

1.2 Spin, aim, and feed

The current stub has no physical feeder/servo model. Its downward record is `RobotAction(motors={}, servos={}, events=[Event("shooter.feed.start", t, {})])` then a matching `shooter.feed.end`; a future real subsystem would own any servo pulse.

1.3 Action, HAL, and completion feedback

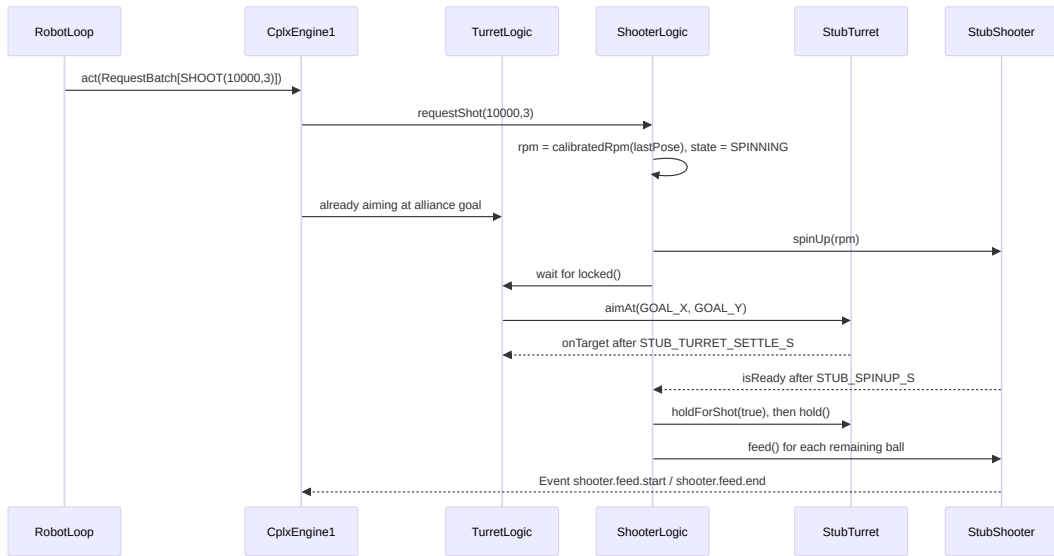


Figure 5: Diagram 5

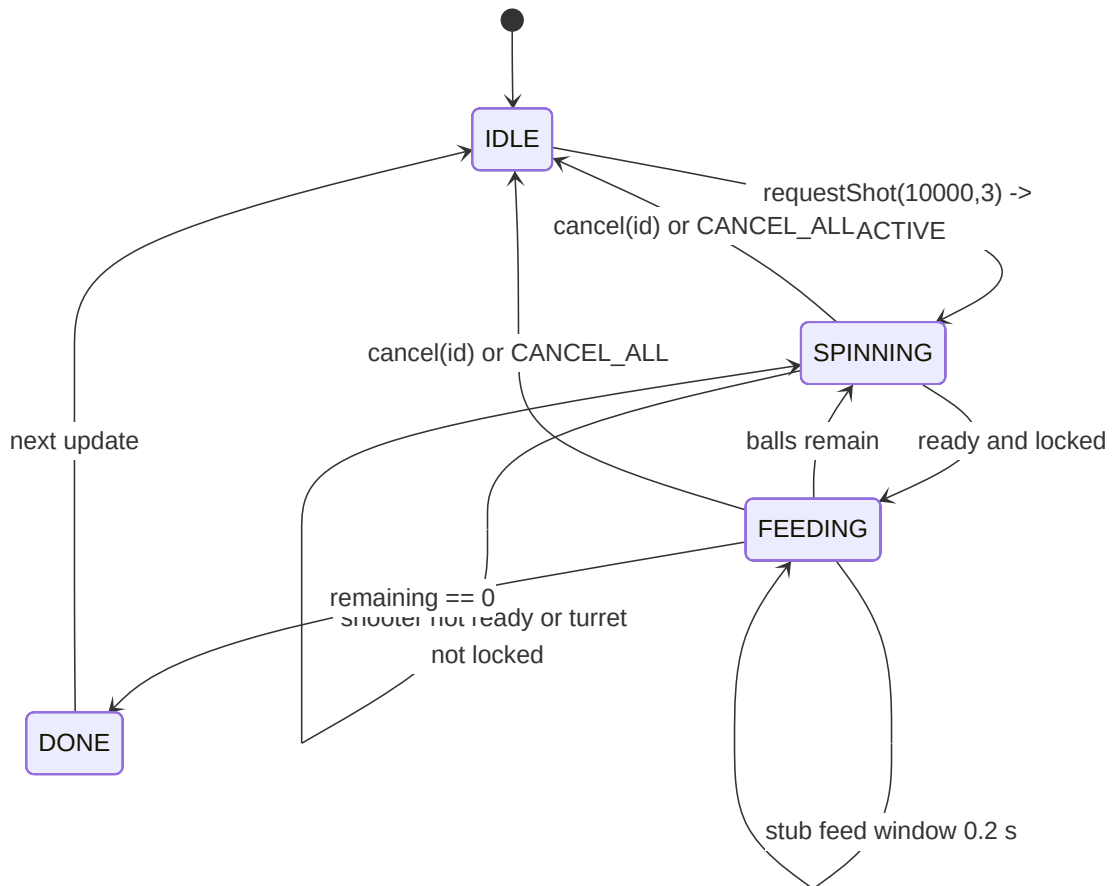


Figure 6: Diagram 6

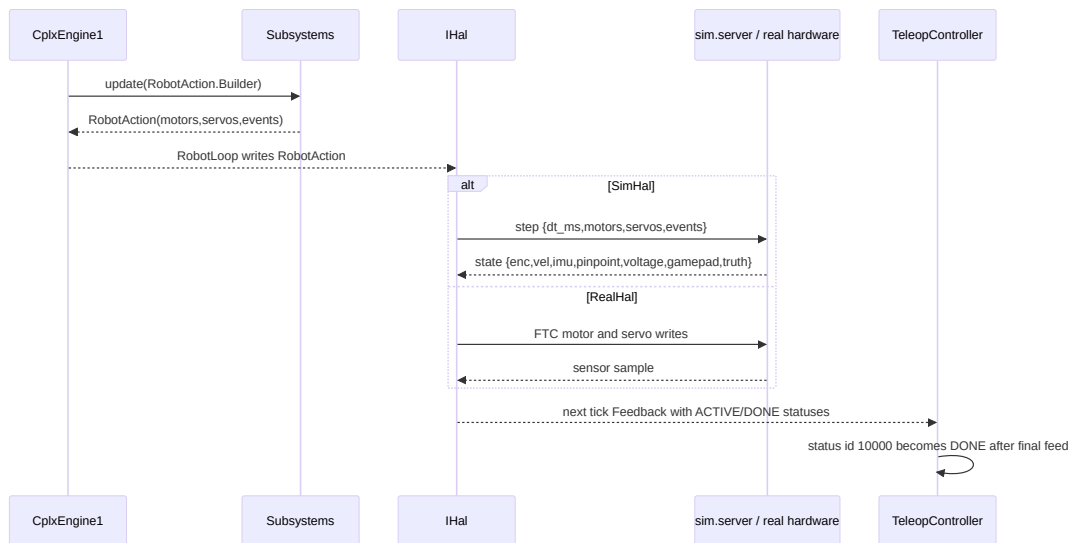


Figure 7: Diagram 7

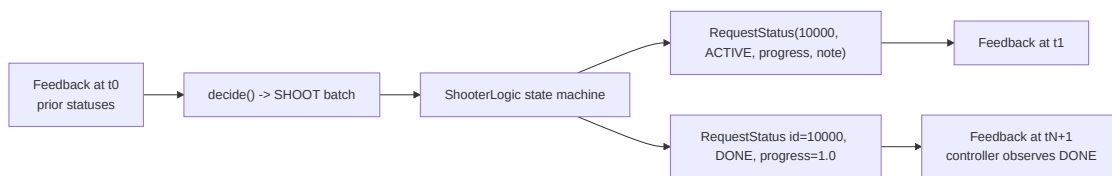


Figure 8: Diagram 8

2. GOTO(path) / PATH from auto to drive completion

2.1 Auto builder emits the path record

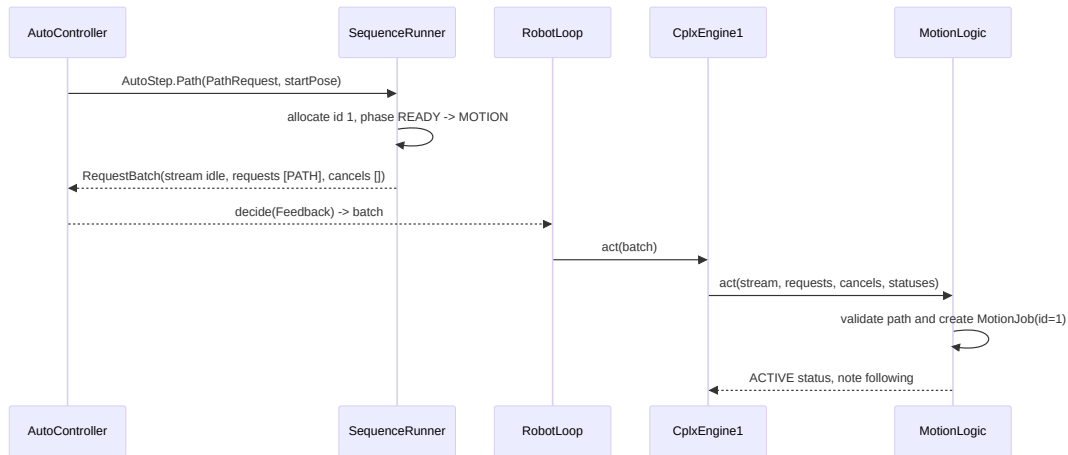


Figure 9: Diagram 9

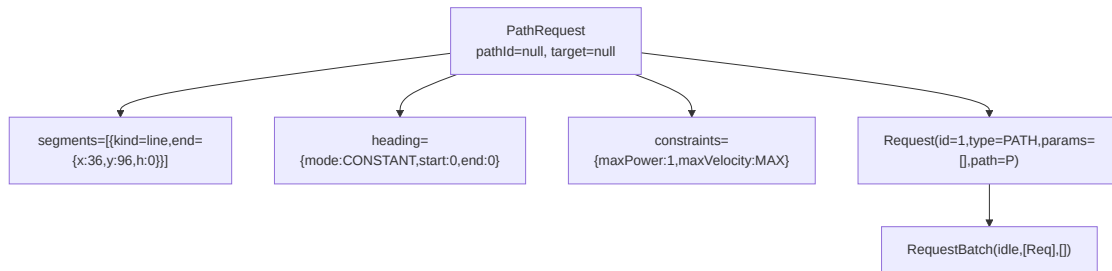


Figure 10: Diagram 10

The canonical logic-seam JSON uses these exact keys (from `SeamJson.path`):

```
{"seam":"logic","t_ms":1000,"batch":{"stream":{"vx":0.0,"vy":0.0,"omega":0.0,"manualDrive":false},"requests":[]}}
```

2.2 Pedro, action, and DONE

GOTO is the three-number request form (`[x,y,headingRad]`) consumed by the same `MotionLogic.start` path; an auto builder normally emits PATH with a `PathRequest` instead. A busy drive or malformed path is REJECTED, never guessed.

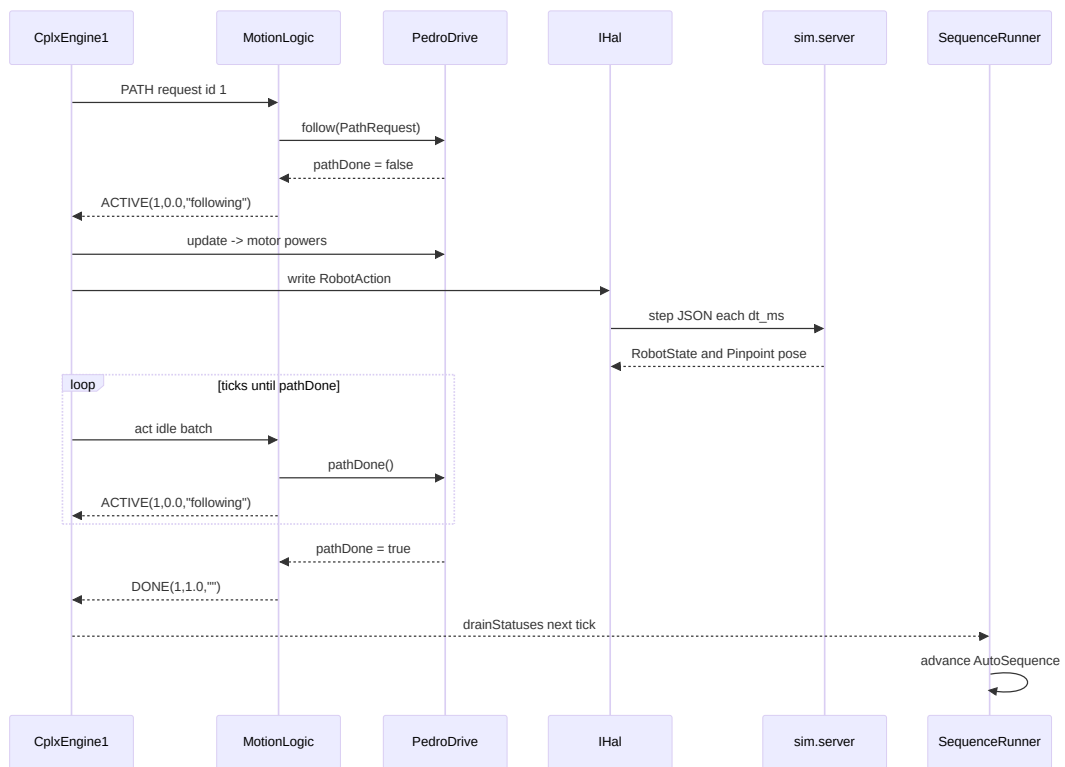


Figure 11: Diagram 11

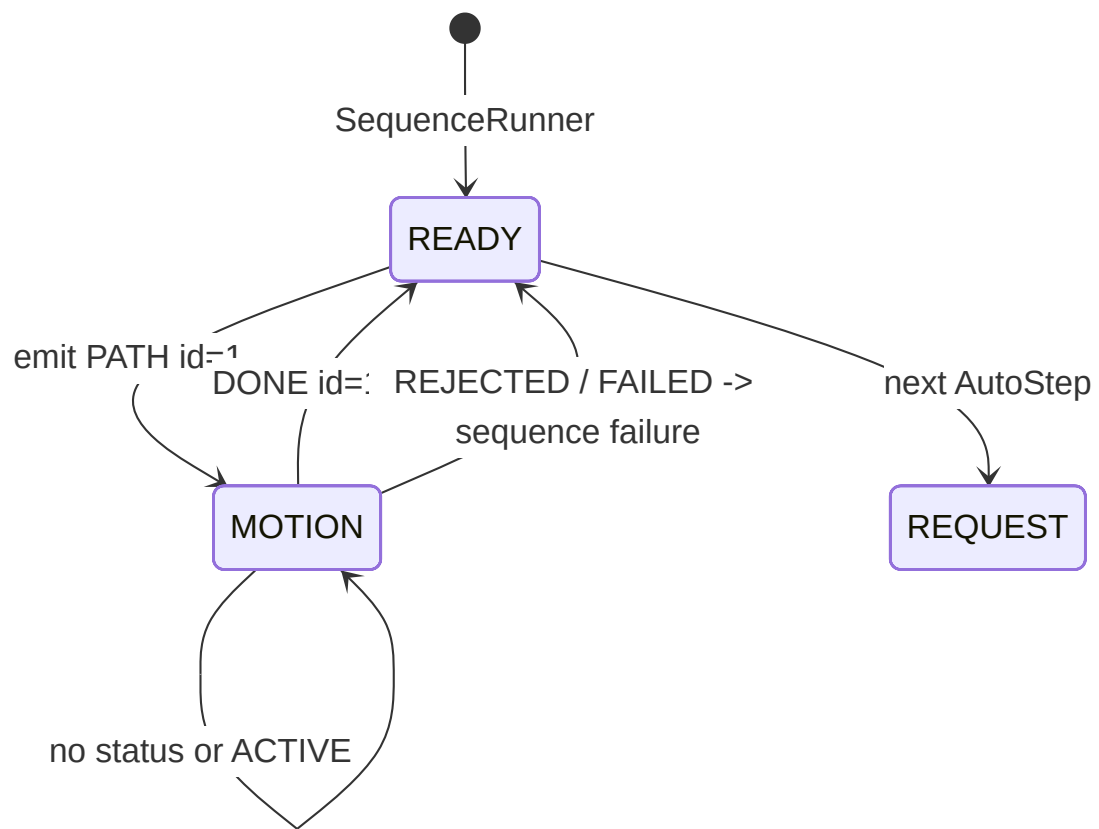


Figure 12: Diagram 12

3. SWITCH_ENGINE at a loop boundary

3.1 START gesture to selection

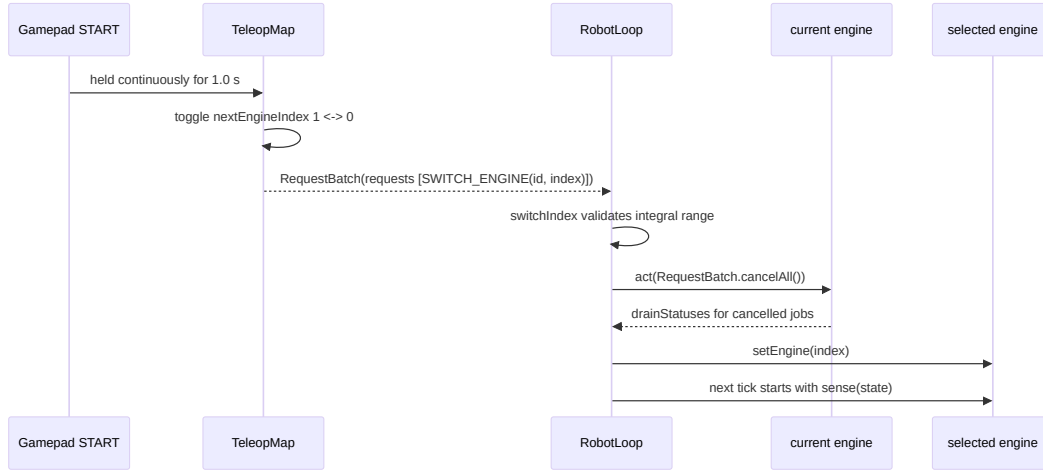


Figure 13: Diagram 13



Figure 14: Diagram 14

3.2 Exact handoff records and feedback

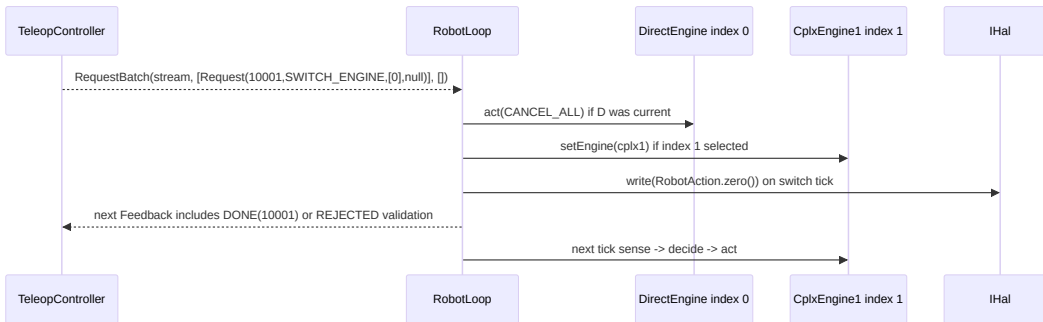


Figure 15: Diagram 15

The factory order is index 0 = direct, 1 = cplx1. SWITCH_ENGINE never reaches DirectMap or a cplx module; RobotLoop strips it, and statuses are visible upward one tick later.

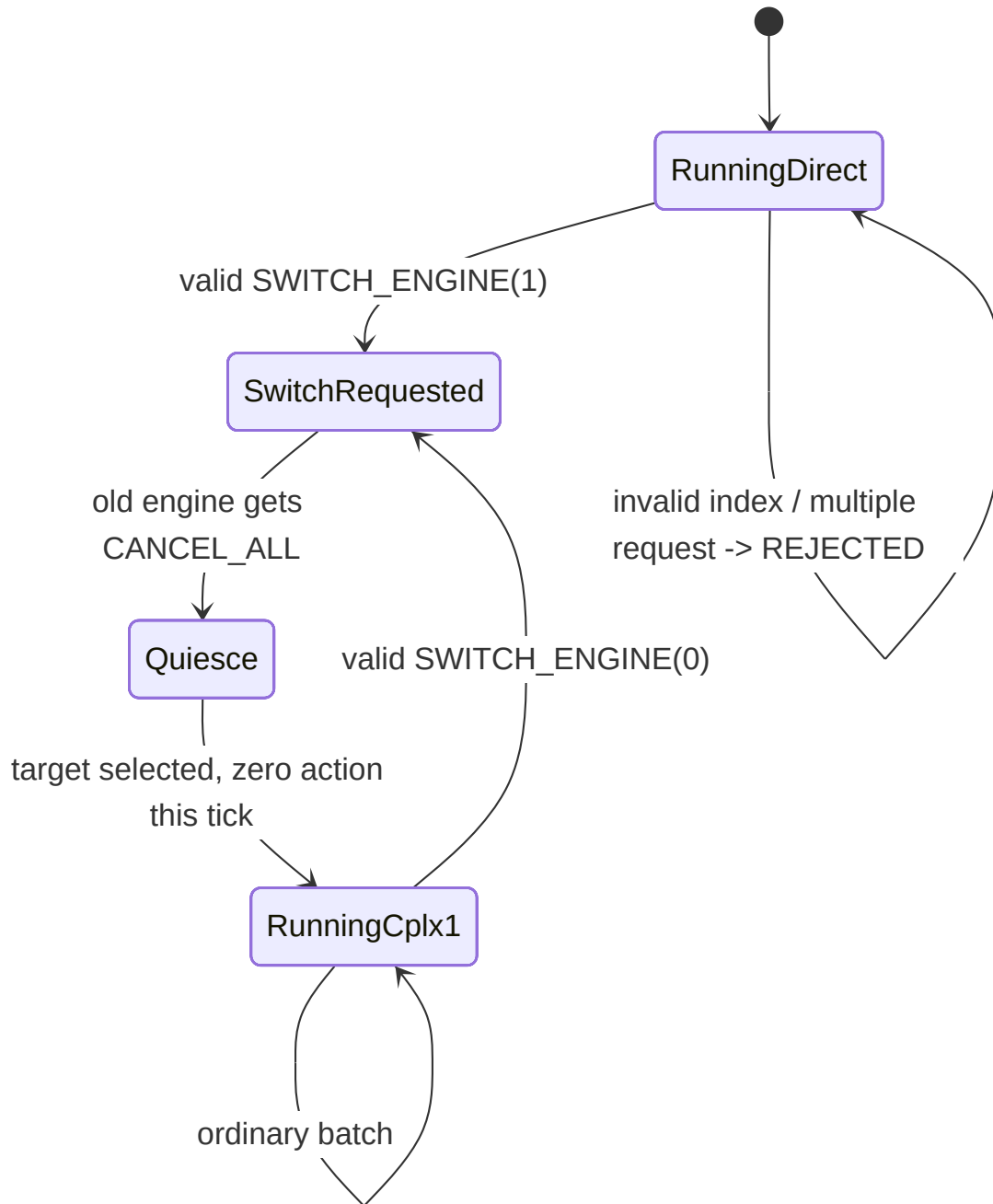


Figure 16: Diagram 16

4. The same seams in JSONL

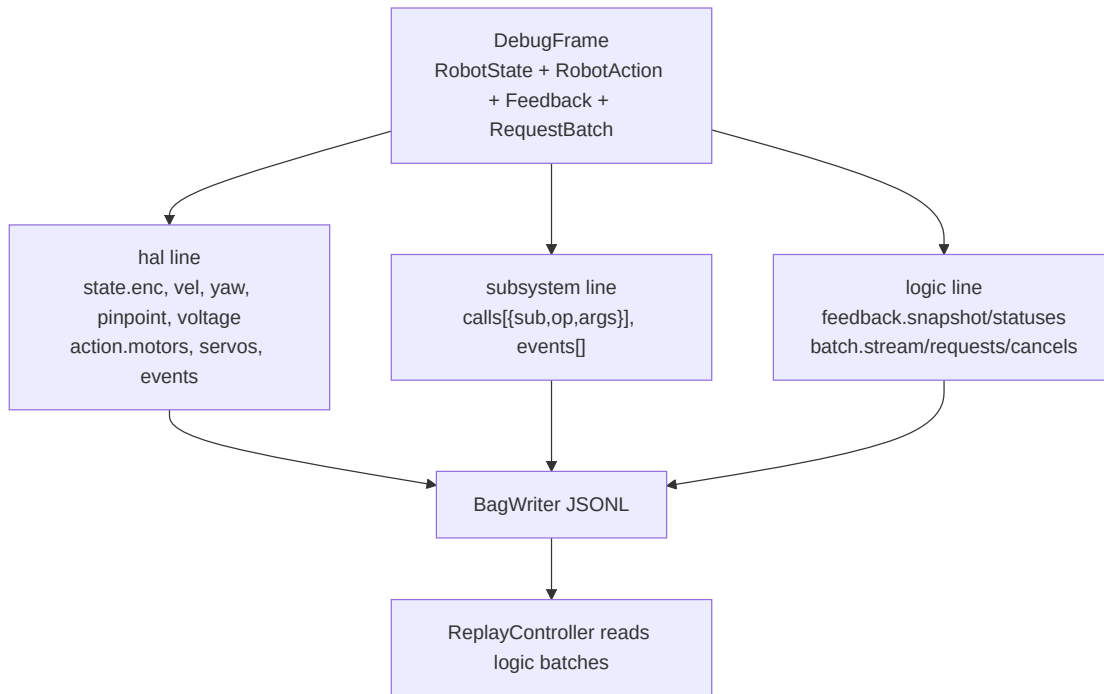


Figure 17: Diagram 17

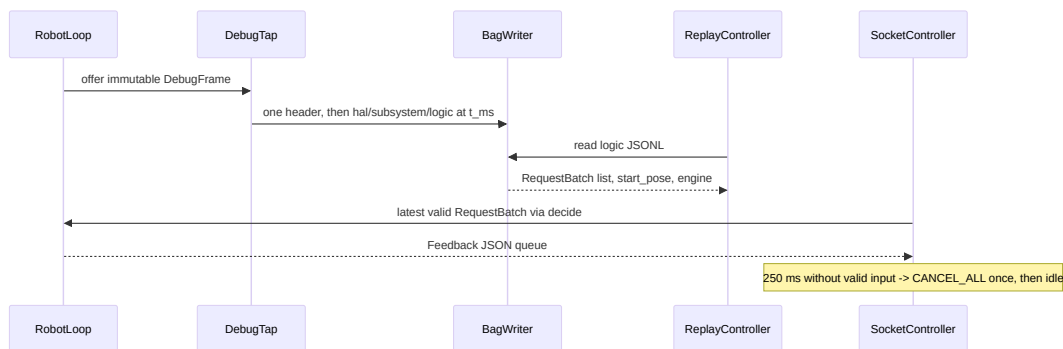


Figure 18: Diagram 18

RobotState never carries simulator **truth**; it is retained only by **SimHal** for tests/viewer. The record path is the same on the real robot and simulator; only **IHal** changes.